

**DEPARTMENT OF COMPUTER SCIENCE ENGINEERING
LAB PRACTICAL FILE**



PROBABILITY, STATISTICS AND LINEAR PROGRAMMING LAB

BS-252

DR. AKHILESH DAS GUPTA INSTITUTE OF PROFESSIONAL STUDIES
NEW DELHI

AFFILIATED TO
GURU GOBIND SINGH INDRAPRASTHA UNIVERSITY

Submitted To: Mr.Yogesh Sharma

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Name: _____

Branch: _____

Section: _____

Enrollment No.: _____

Vision of Dr. Akhilesh Das Gupta Institute of Professional Studies :

To produce globally competent and socially responsible technocrats and entrepreneurs who can develop innovative solutions to meet the challenges of 21st century.

Mission of Dr. Akhilesh Das Gupta Institute of Professional Studies :

M1. To Provide Value-Based Education through Multi Grade Teaching Methodologies and Modern Education Facilities.

M2. To Sustain an Active Partnership Program with Industry and Other Academic Institutes with an Aim to Promote Knowledge and Resource Sharing.

M3. To Conduct Value-Added Training Programme to Enhance Employability.

M4. To Provide a Conducive Environment for Development of Ethical and Socially Responsible Technocrats, Managers and Entrepreneurs.

Vision of Department of Computer Science Engineering :

To produce competent computer science engineers having distinct employability skills, involving innovative ideas to fulfil societal needs.

Mission of Department of Computer Science Engineering :

M1. To provide resourceful education through training and skill upgradation.

M2. To inspire the young dynamic minds towards innovation and research to meet the societal needs and responsibilities.

M3. To strengthen the industry-academia interface for better employability.

Program: -1

Aim :- Installation of Scilab and demonstration of simple programming concepts like matrix multiplication (scaler and vector), loop, conditional statements and plotting.

i.

```
>> a = [ 1 2 3 ; 4 5 6; 7 8 9];  
b = [ 7 5 6 ; 2 0 8; 5 7 1];  
c = a + b  
d = a - b
```

```
c =
```

```
      8      7      9  
      6      5     14  
     12     15     10
```

```
d =
```

```
     -6     -3     -3  
      2      5     -2  
      2      1      8
```

```
>> e= a*b
```

```
e =
```

```
     26     26     25  
     68     62     70  
    110     98    115
```

```
>> f=a/b
```

```
f =
```

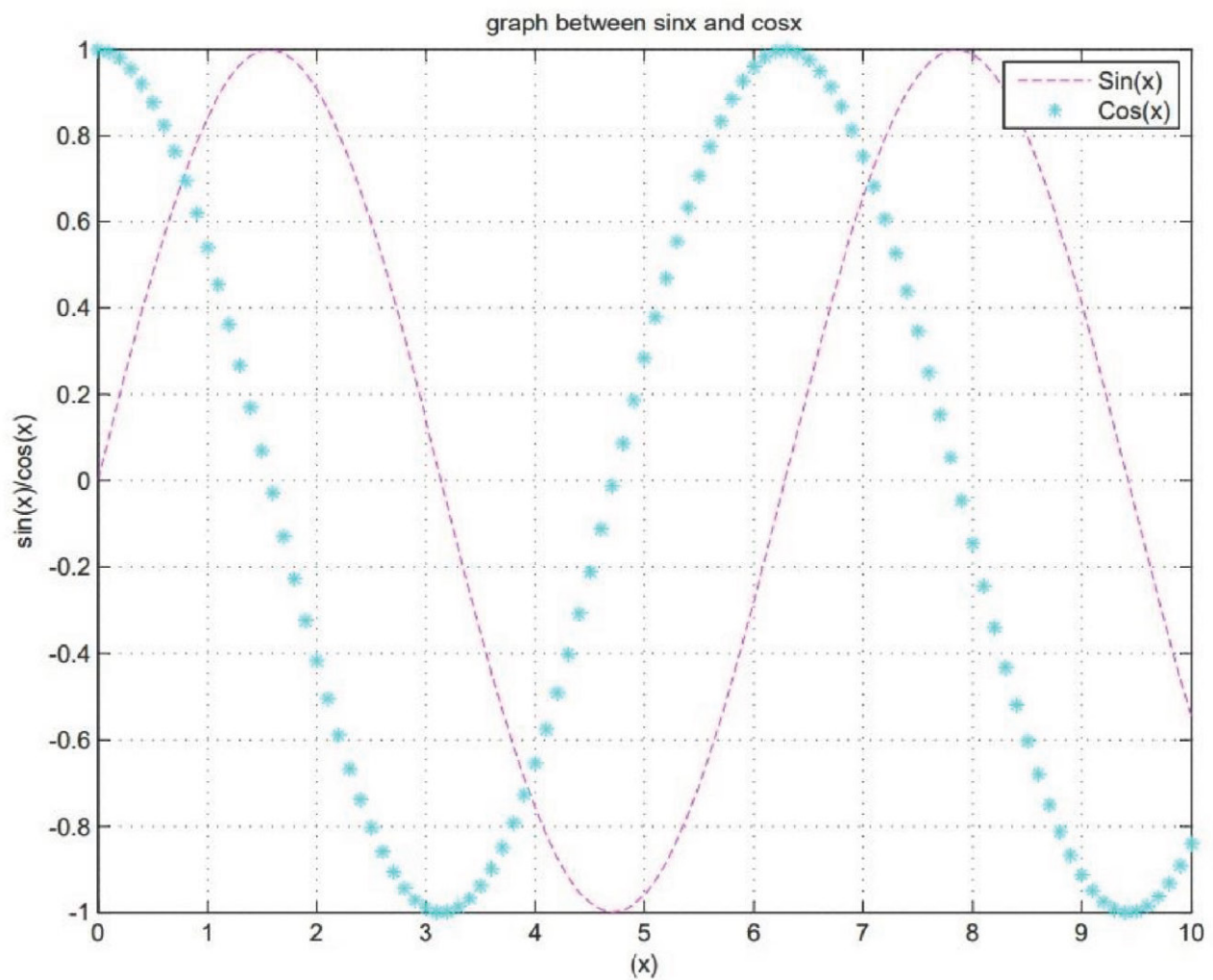
```
   -0.5254    0.6864    0.6610  
   -0.4237    0.9407    1.0169  
   -0.3220    1.1949    1.3729
```

```
>>
```

ii.

```
>> x = [0 : 0.1: 10];  
y = sin(x);  
g = cos(x);  
plot(x, y, 'm--', x, g, 'c*'), xlabel('(x)'), ylabel('sin(x)/cos(x)'), title('graph between  
sinx and cosx'), legend('Sin(x)', 'Cos(x)'), grid on;  
>>
```

Output:-



Program:- 2

Aim :- Fitting of Binomial Distribution for given n and p.

a) Out of n trials k are successes

```
>> q=0.25;  
p=1-q;  
n=6;  
x=3;  
prob=nchoosek(n,x) * (p^x) * (q^(n-x));disp(prob)
```

Output:-

```
0.1318
```

b) kth success occurs at the nth trial

```
>> q=0.25;  
p=1-q;  
n=8;  
x=4;  
prob=nchoosek(n,x) * (p^x) * p * (q^(n-x));disp(prob)
```

Output:-

```
0.0649
```

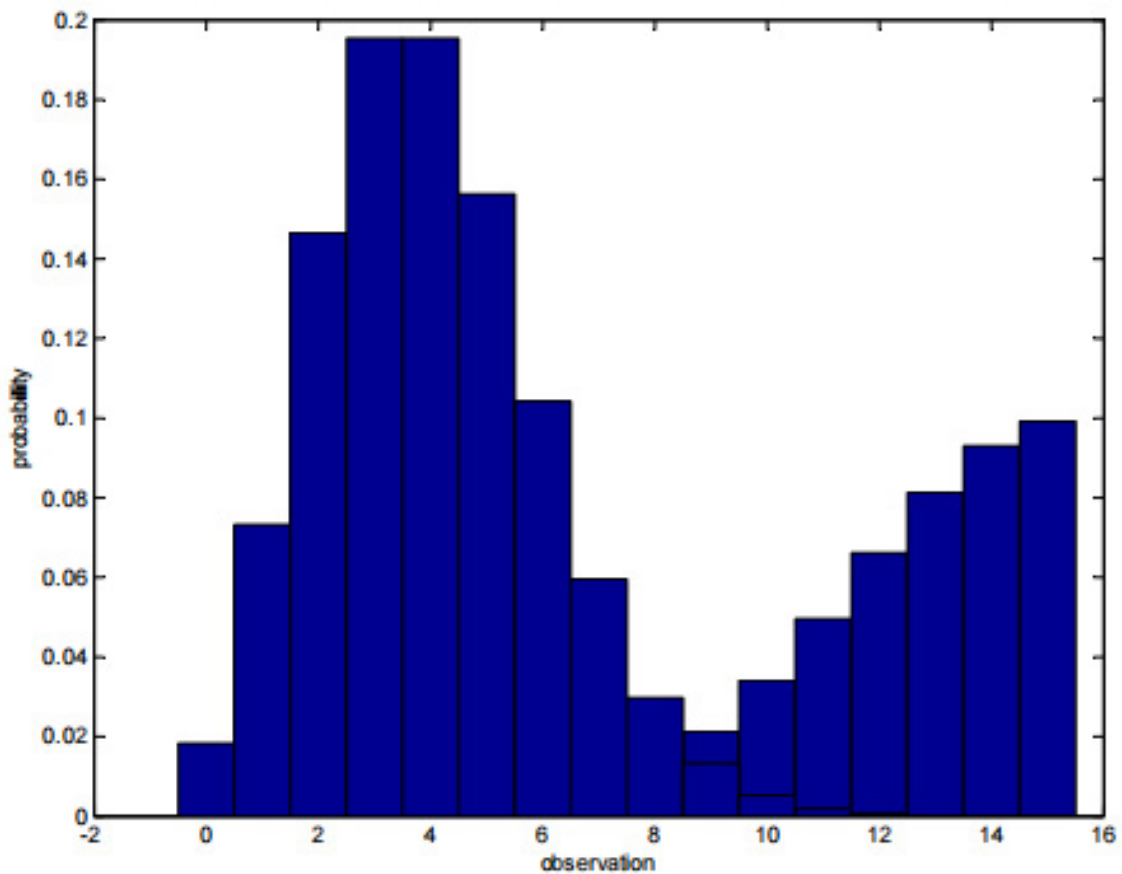
Program: -3

Aim :- Fitting of Poisson Distribution of given value of Lamda.

i.

```
>> %compute probability distribution function
>> x=0:15;
y=poisspdf(x,4);
bar(x,y,1)
xlabel('observation')
ylabel('probability')
```

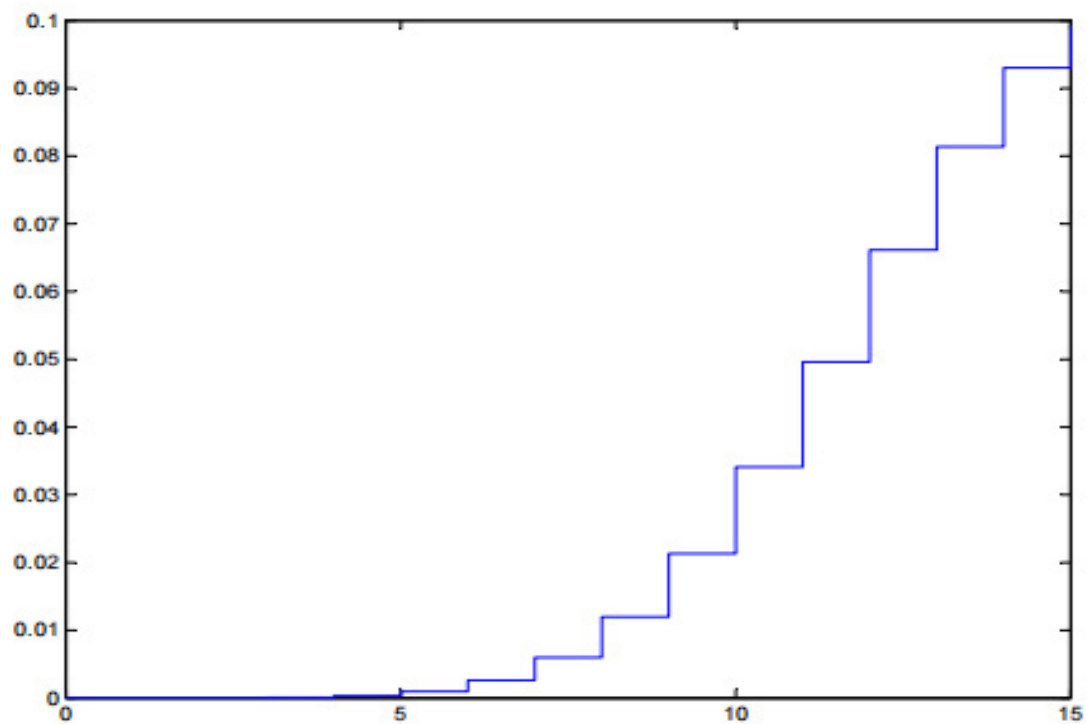
Output:-



ii.

```
>> %compute cumulative probability distributive function  
>> stairs(x,y)  
>> xlabel('observation')  
>> ylabel('cumulative probability')
```

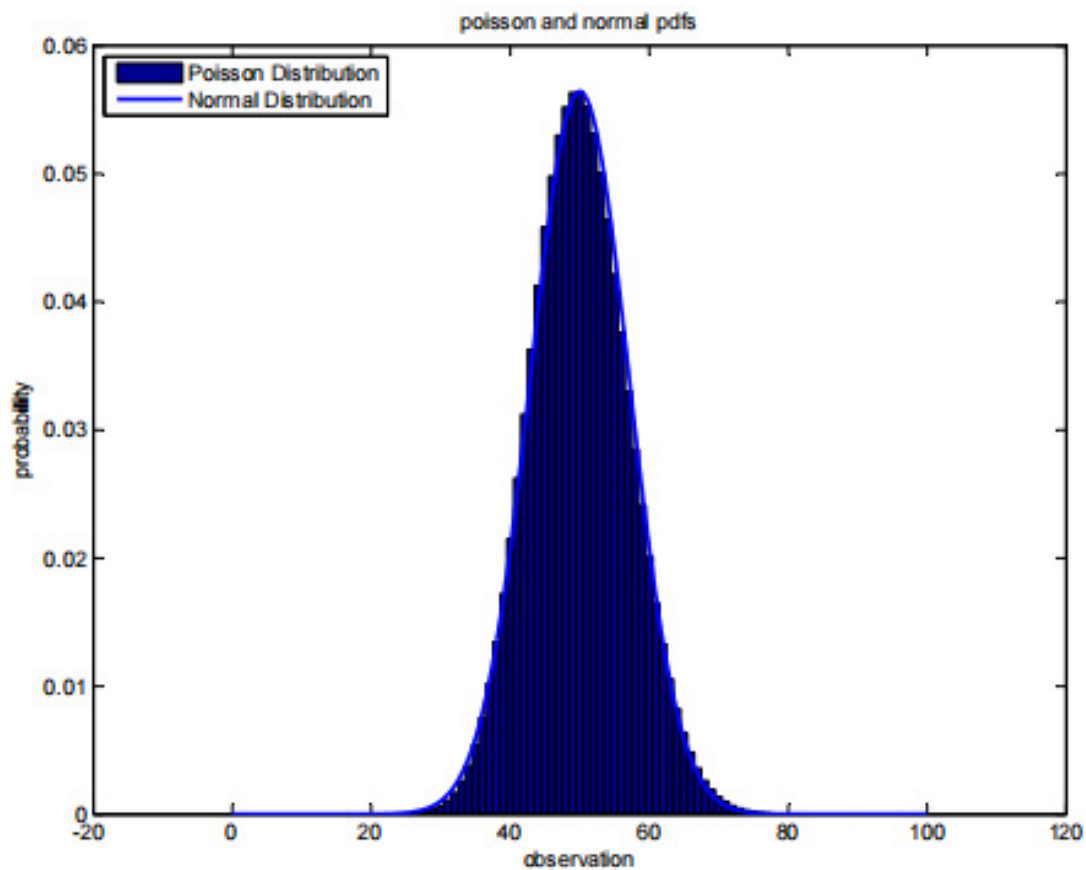
Output:-



iii.

```
>> %Compare Normal and Poisson Distribution Function
>> lamda=50;
x1=0:100;
y1=poisspdf(x1, lamda);
mu=lamda;
sigma=sqrt(lamda);
x2=0:0.1:100;
y2=normpdf(x2, mu, sigma);
bar(x1, y1, 1)
hold on
plot(x2, y2, 'LineWidth', 2)
xlabel('observation')
ylabel('probability')
title('poisson and normal pdfs')
legend('Poisson Distribution', 'Normal Distribution', 'Location', 'northwest')
hold off
```

Output:-



Program: - 4

Aim :- Fitting of Normal Distribution when parameters are given.

% Fit Normal Distribution Object

```
>>load examgrades
```

```
>>x = grades(:,1);
```

```
>>pd = fitdist(x,'Normal')
```

```
pd = NormalDistribution
```

```
Normal distribution
```

```
mu = 75.0083 [73.4321, 76.5846]
```

```
sigma = 8.7202 [7.7391, 9.98843]
```

% Estimate Parameters

```
>>load examgrades
```

```
>>x = grades(:,1);
```

```
>>[mu,s,muci,sci] = normfit(x)
```

```
mu = 75.0083
```

```
s = 8.7202
```

```
muci = 2×1
```

```
73.4321
```

```
76.5846
```

```
sci = 2×1
```

```
7.7391
```

```
9.9884
```

```
>>s^2
```

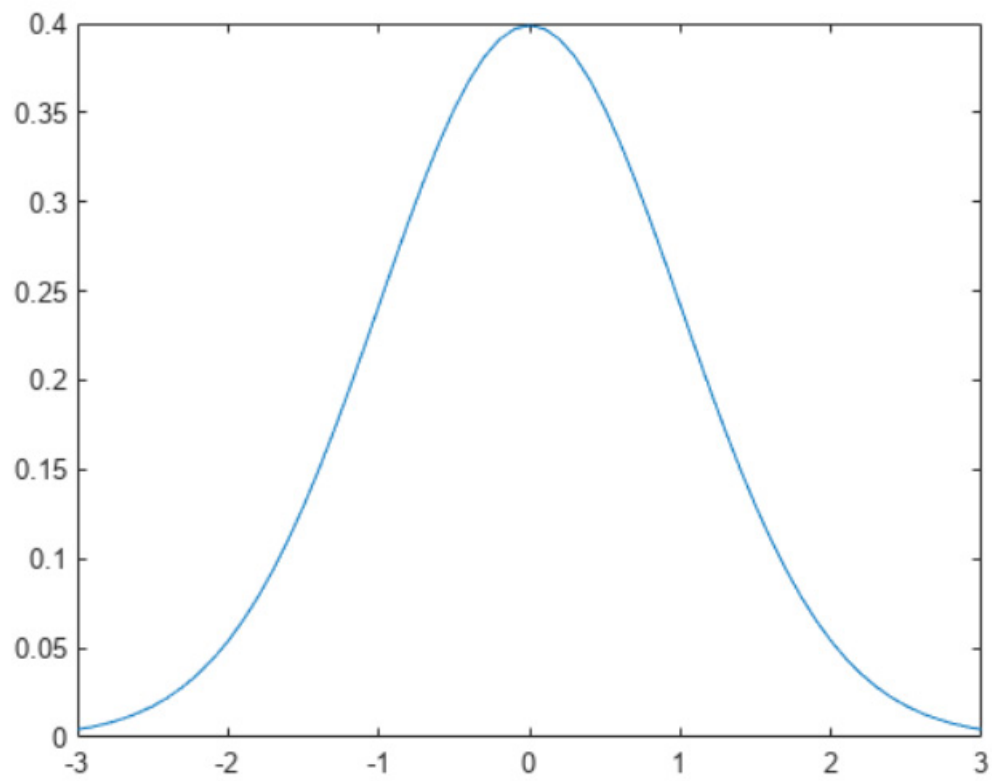
```
ans = 76.0419
```

%Compute and Plot the Normal Distribution pdf

```
>>x = [-3:.1:3];
```

```
>>y = normpdf(x,0,1);
```

```
>>plot(x,y)
```



%Plot Standard Normal Distribution cdf

```
>>pd = makedist('Normal')
```

```
pd =
```

```
NormalDistribution
```

```
Normal distribution
```

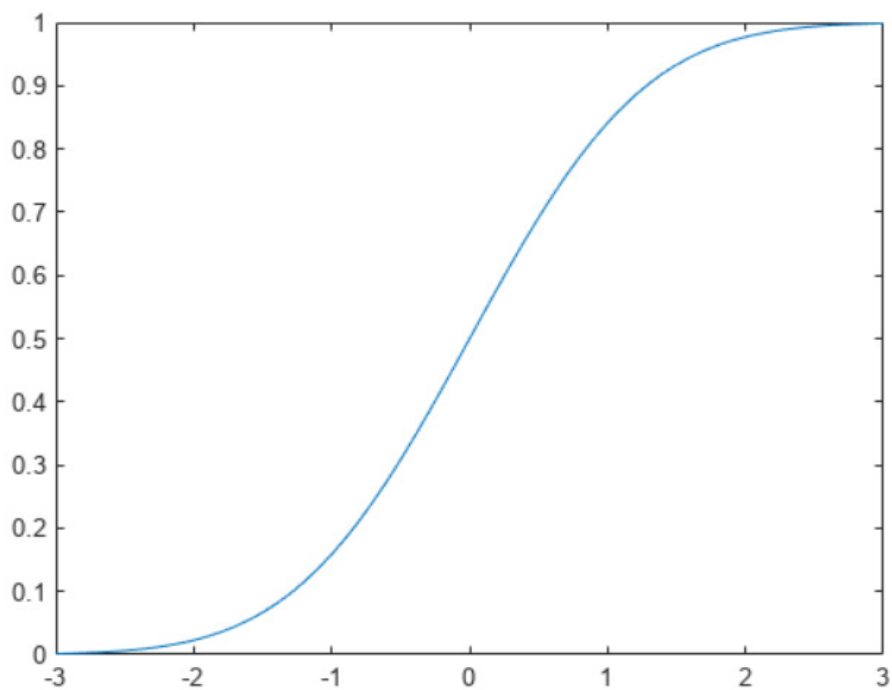
```
mu = 0
```

```
sigma = 1
```

```
>>x = -3:.1:3;
```

```
>>p = cdf(pd,x);
```

```
>>plot(x,p)
```



%Compare Gamma and Normal Distribution pdfs

```
>>a = 100;  
>>b = 5;  
>>x = 250:750;  
>>y_gam = gampdf(x,a,b);
```

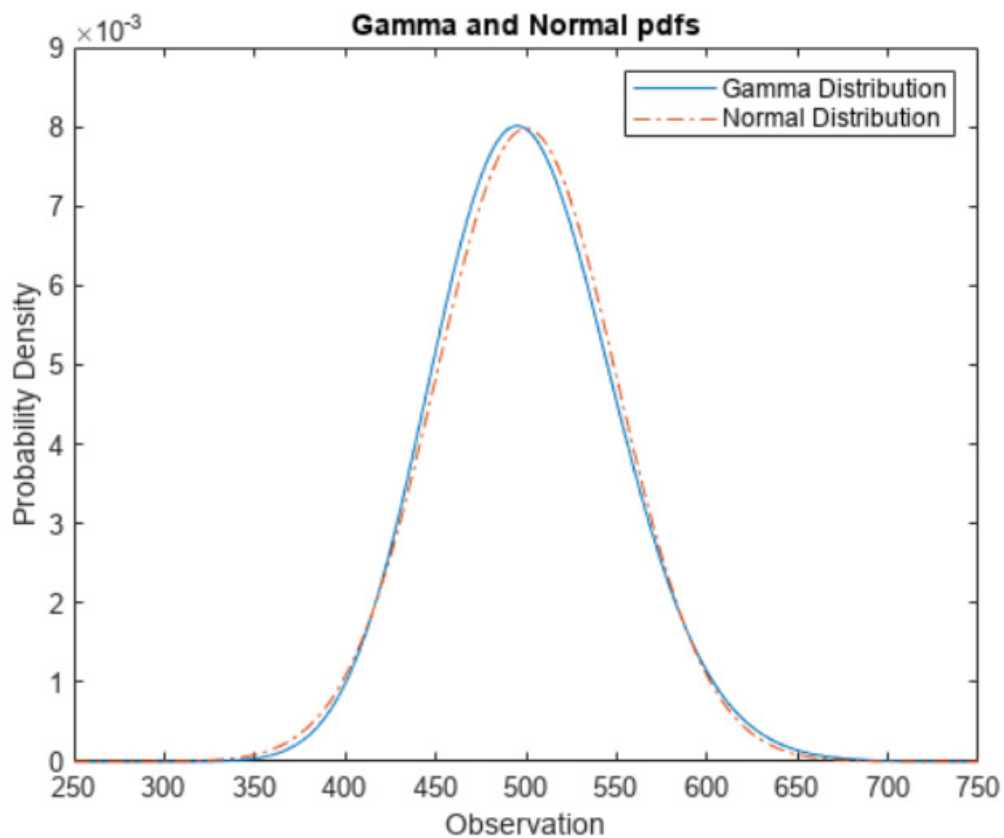
```
>>mu = a*b
```

```
mu = 500
```

```
>>sigma = sqrt(a*b^2)
```

```
sigma = 50
```

```
>>y_norm = normpdf(x,mu,sigma);  
>>plot(x,y_gam,'-',x,y_norm,'-')  
>>title('Gamma and Normal pdfs')  
>>xlabel('Observation')  
>>ylabel('Probability Density')  
>>legend('Gamma Distribution','Normal Distribution')
```



%Relationship Between Normal and Lognormal Distributions

```
>>pd = makedist('Lognormal','mu',5,'sigma',2)
```

```
pd =
```

```
LognormalDistribution
```

```
Lognormal distribution
```

```
mu = 5
```

```
sigma = 2
```

```
>>mean(pd)
```

```
ans = 1.0966e+03
```

```
>>rng('default'); % For reproducibility
```

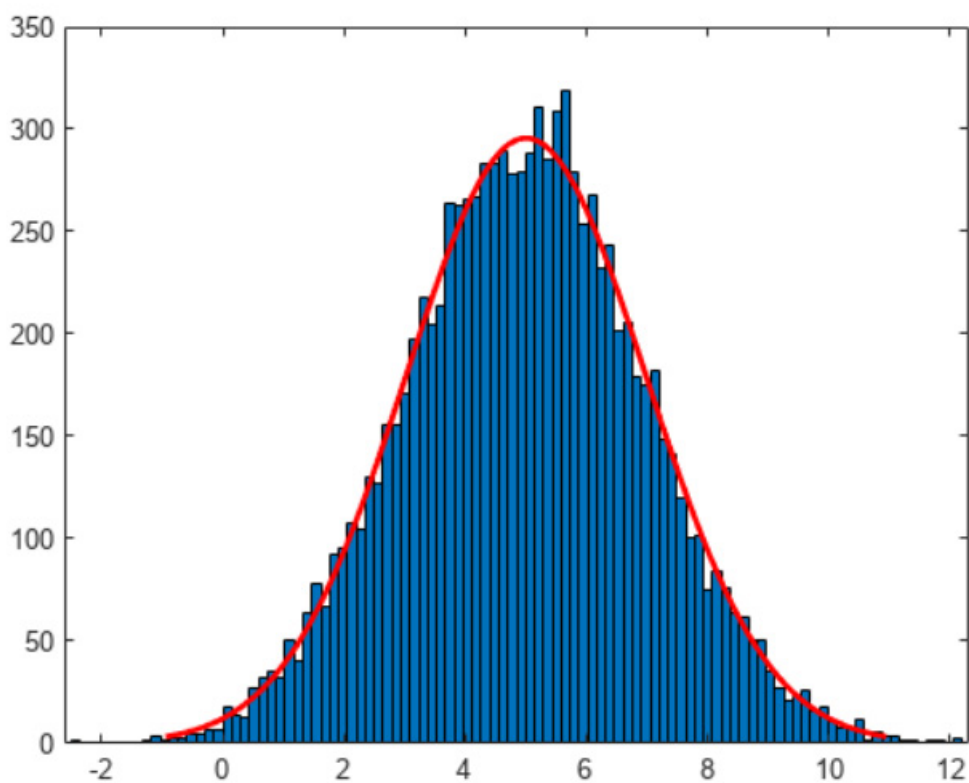
```
>>x = random(pd,10000,1);
```

```
>>logx = log(x);
```

```
>>m = mean(logx)
```

```
m = 5.0033
```

```
>>histfit(logx)
```



```
>>pd_normal = fitdist(logx,'Normal')
```

```
pd_normal =
```

```
NormalDistribution
```

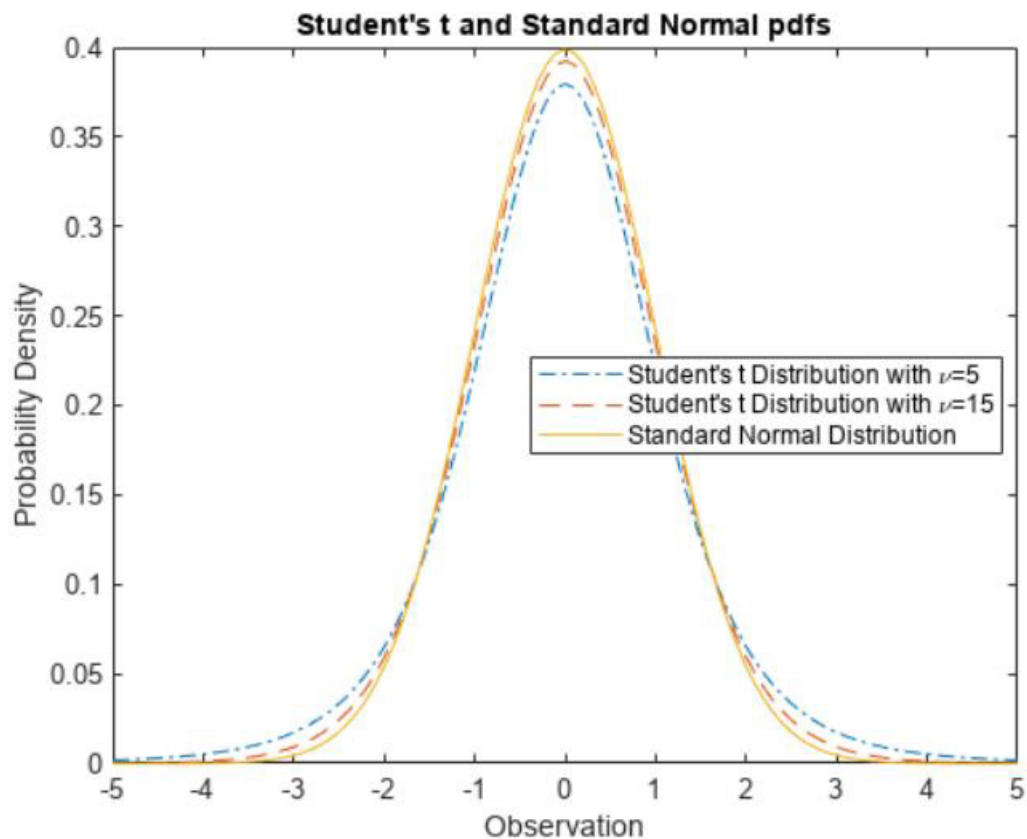
```
Normal distribution
```

```
mu = 5.00332 [4.96445, 5.04219]
```

```
sigma = 1.98296 [1.95585, 2.01083]
```

%Compare Student's t and Normal Distribution pdfs

```
>>x = [-5:0.1:5];  
>>y1 = tpdf(x,5);  
>>y2 = tpdf(x,15);  
>>z = normpdf(x,0,1);  
>>plot(x,y1,'-!',x,y2,'--!',x,z,'-')  
>>legend('Student's t Distribution with \nu=5', ...  
        'Student's t Distribution with \nu=15', ...  
        'Standard Normal Distribution','Location','best')  
>>xlabel('Observation')  
>>ylabel('Probability Density')  
>>title('Student's t and Standard Normal pdfs')
```



Program: - 5

Aim :- Program to plot exponential distribution for various parametric values.

%Fit Exponential Distribution to Data

```
>>x = exprnd(700,100,1); % Generate sample
```

```
>>pd = fitdist(x,'exponential')
```

```
pd =
```

```
ExponentialDistribution
```

```
Exponential distribution
```

```
mu = 641.934 [532.598, 788.966]
```

```
>>[muhat,muci] = expfit(x) % Distribution specific function
```

```
muhat = 641.9342
```

```
muci = 2×1
```

```
532.5976
```

```
788.9660
```

```
>>[muhat2,muci2] = mle(x,'distribution','exponential') % Generic distribution function
```

```
muhat2 = 641.9342
```

```
muci2 = 2×1
```

```
532.5976
```

```
788.9660
```

%Compute Exponential Distribution pdf

```
>>x = 0:0.1:10;
```

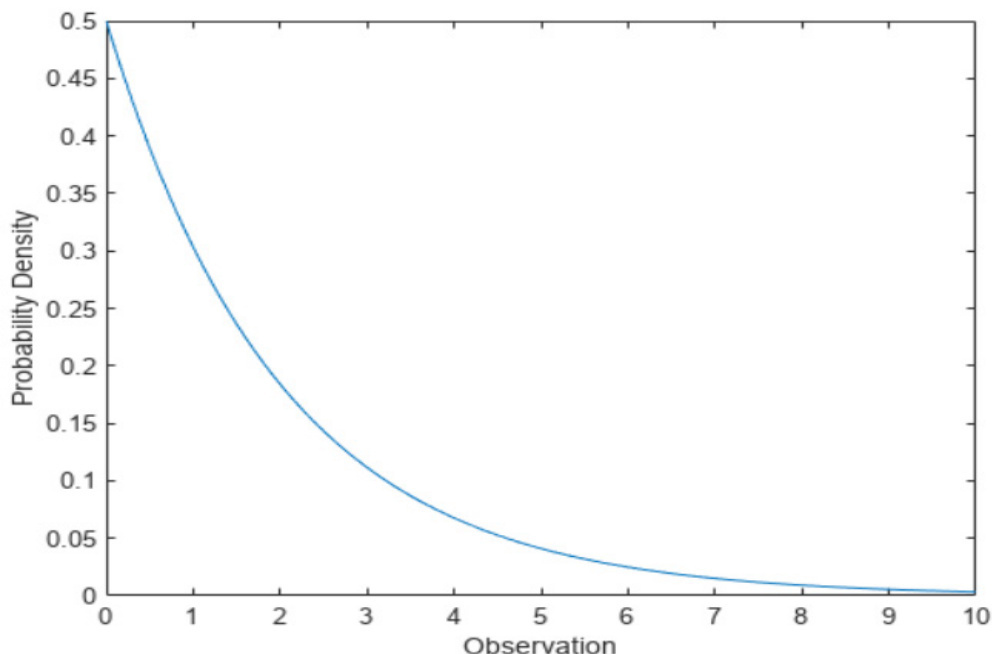
```
>>y = exppdf(x,2);
```

```
>>figure;
```

```
>>plot(x,y)
```

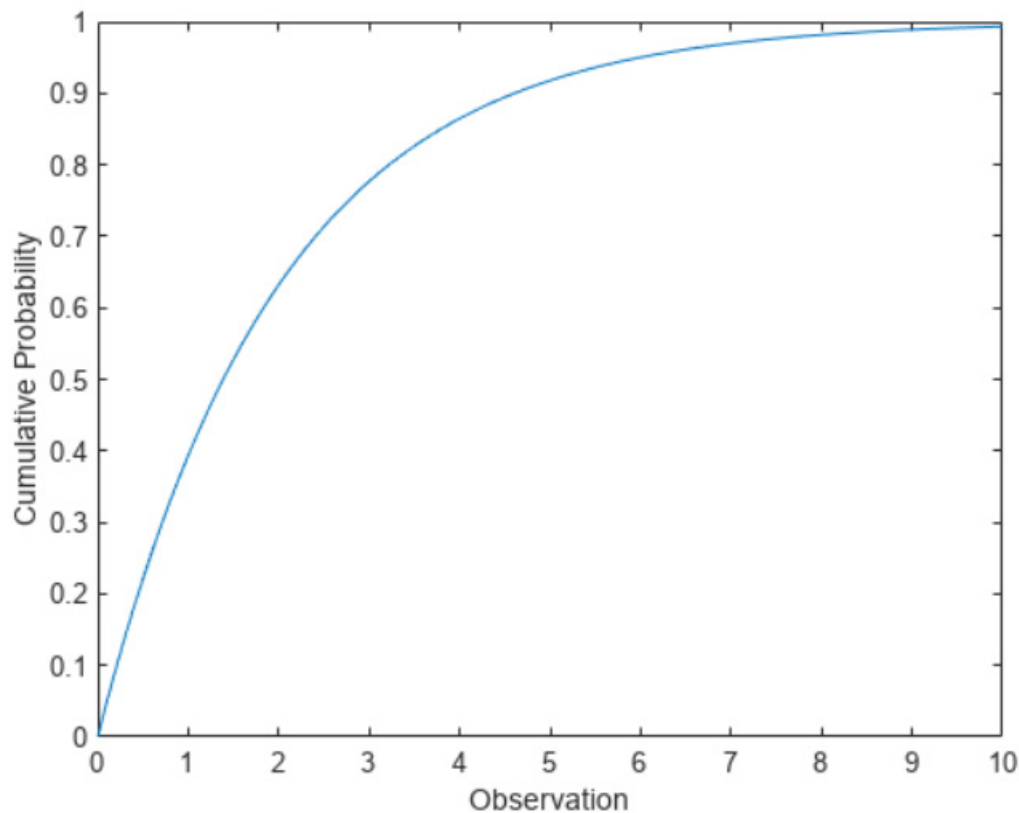
```
>>xlabel('Observation')
```

```
>>ylabel('Probability Density')
```



%Compute Exponential Distribution cdf

```
>>x = 0:0.1:10;  
>>y = expcdf(x,2);  
>>figure;  
>>plot(x,y)  
>>xlabel('Observation')  
>>ylabel('Cumulative Probability')
```



%Exponentially Distributed Lifetimes

```
>>x = 1:5;  
>>lambda1 = exppdf(x,2)./(1-expcdf(x,2))  
lambda1 = 1×5
```

0.5000 0.5000 0.5000 0.5000 0.5000

```
>>mu = 1:5;  
>>lambda2 = exppdf(3,mu)./(1-expcdf(3,mu))  
lambda2 = 1×5
```

1.0000 0.5000 0.3333 0.2500 0.2000

```
>>x2 = 5:5:25;  
>>x3 = x2 + 1;  
>>deltap = (expcdf(x3,10)-expcdf(x2,10))./(1-expcdf(x2,10))  
deltap = 1×5
```

0.0952 0.0952 0.0952 0.0952 0.0952

